

```

dima@HP:~/try_2/453503_VASILENKO_7$ docker run -d -p 80:80 docker/getting-starte
d
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
c158987b0551: Pull complete
4f7e34c2de10: Pull complete
f1d1c9928c82: Pull complete
9b6f639ec6ea: Pull complete
1e35f6679fab: Pull complete
ee68d3549ec8: Pull complete
b6334b6ace34: Pull complete
33e0cbbb4673: Pull complete
cb9626c74200: Pull complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
5f395aaa0b55b2b3ee900709bf24b8d06954e70b7a88c7aff130c9ed3f56a9f4

```

```

Main
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2$ mkdir dock_im
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2$ cd dock_im/
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ touch DockerFile
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ touch main.py
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ vim main.py
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ vim DockerFile
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ vim main.py
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ vim DockerFile
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ vim main.py
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ docker build .
[+] Building 0.2s (1/1) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile             0.1s
=> => transferring dockerfile: 2B                               0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerf
ile: no such file or directory
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ ls

```

```

dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock_im$ docker build .
[+] Building 6.7s (6/8)                                         docker:desktop-linux
=> [internal] load build definition from Dockerfile             0.0s
=> => transferring dockerfile: 210B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 4.2s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/4] FROM docker.io/library/python:3.11-slim@sha256:c8271b1f627d0068 0.0s
=> => resolve docker.io/library/python:3.11-slim@sha256:c8271b1f627d0068 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 29B                                   0.0s
=> CACHED [2/4] WORKDIR /app                                    0.0s
=> [3/4] RUN pip install https://github.com/sergeymakarov/geometric-lib-si 2.4s

```

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
app:latest	11cbf4d9bc65	379MB	101MB	
docker/getting-started:latest	d79336f4812b	73.9MB	21.4MB	U
docker/welcome-to-docker:latest	c4d56c24da4f	22.2MB	6.03MB	

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/dock$ docker run app
```

Square area: 78.53981633974483
Rectangle area: 1.0

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/node-express-server$ docker compose build backend --no-cache
```

docker compose up -d
WARN[0000] /home/dima/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/docker-compose.yaml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Building 55:15 (13/13) FINISHED
=> [internal] load local bake definitions 0.0s
=> == reading from stdin 754B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> == transferring dockerfile: 336B 0.0s
=> [internal] load metadata for docker.io/library/node:lts-alpine 1.0s
=> [internal] load .dockerignore 0.0s
=> == transferring context: 2B 0.0s
=> [build 1/5] FROM docker.io/library/node:lts-alpine@sha256:7fddd9ddeae8196abf4a3ef2de34e11f7b1a722119f91f28ddf1e99dcafd114 0.0s
=> == resolve docker.io/library/node:lts-alpine@sha256:7fddd9ddeae8196abf4a3ef2de34e11f7b1a722119f91f28ddf1e99dcafd114 0.0s
=> [internal] load build context 0.0s
=> == transferring context: 535B 0.0s
=> CACHED [build 2/5] WORKDIR /app 0.0s
=> [build 3/5] COPY package*.json ./ 0.0s
=> [build 4/5] RUN npm install 49.2s

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/node-express-server$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
angular-15-node-js-mongodb-example-database-1	mongo	"docker-entrypoint.s..."	database	2 hours ago	Up 2 hours	0.0.0.0:27017->27017/tcp
angular-15-node-js-mongodb-example-frontend-1	angular-15-node-js-mongodb-example-frontend	"/docker-entrypoint..."	frontend	2 hours ago	Up 2 hours	0.0.0.0:4200->80/tcp

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/node-express-server$ docker compose ps
```

NAME	IMAGE	COMMAND	SERVICE	CREATED
backend	angular-15-node-js-mongodb-example-backend	"docker-entrypoint.s..."	backend	6 minutes ago
database	mongo	"docker-entrypoint.s..."	database	6 minutes ago
frontend	angular-15-node-js-mongodb-example-frontend	"/docker-entrypoint..."	frontend	6 minutes ago

Информация о сетях:

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/node-express-server$ docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
1818e6488f10	bridge	bridge	local
311a9e17d237	host	host	local
7f82da57511f	none	null	local
aa07e0fc2f6c	smarttrailsbackend_default	bridge	local

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/node-express-server$ docker network inspect bridge
[
  {
    "Name": "bridge",
    "Id": "1818e6488f101def0ceacff817419d5e6e2c7e2e479cad5eefa06bff4475fad9",
    "Created": "2026-03-01T22:24:35.046887544+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "Gateway": "172.17.0.1"
        }
      ]
    }
  }
]
```

Создаю свою сеть:

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/node-express-server$ docker network create my-bridge
8921180b0e36c4b18771c11afc4ca9cb14ac8731b6afe1bedd0e5c10c0424dea
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/node-express-server$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
1818e6488f10        bridge              bridge              local
311a9e17d237        host                host                local
8921180b0e36        my-bridge           bridge              local
7f82da57511f        none                null                local
aa07e0fc2f6c        smarttrailsbackend_default bridge              local
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/docker_proj/angular-15-node-js-mongodb-example/node-express-server$
```

запуск котнейнера в сети созданной:

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_image$ docker run -d --name my-container --network my-bridge my-image:latest
b8ba5b2b7b36905e5ed738cf13326d0fad23f13b4c6c043da9e7dc40233a1546
```

проверка сотоояния сети:

```

    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
      "e4a05867af6667e1fc41b30c6e0f3a4e71ec724e8ef12877e3fe549d18fe9db0": {
        "Name": "my-container",
        "EndpointID": "d9d23b94bb6be4774c13fdcfc738ac6af3d12ad88445e7e5a907506f869c33b",
        "MacAddress": "d6:cc:67:6c:55:d9",
        "IPv4Address": "172.19.0.2/16",
        "IPv6Address": ""
      }
    },
    "Status": {
      "IPAM": {
        "Subnets": {
          "172.19.0.0/16": {

```

отключаемся от сети контейнером:

```

dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_image$ docker network disconnect my-bridge my-container
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_image$ docker network inspect my-bridge
[
  {
    "Name": "my-bridge",
    "Id": "8921180b0e36c4b18771c11afc4ca9cb14ac8731b6afe1bedd0e5c10c0424dea",
    "Created": "2026-03-02T18:48:19.115611531+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    }
  }
]

```

Создаём ещё одну сеть:

```

dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_image$ docker network create another-bridge
c5359aaa6d71345f22a3a952f13a6f6a56c0f8f3aedc0f8919028be5cfebd177
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_image$ docker network inspect another-bridge
[
  {
    "Name": "another-bridge",
    "Id": "c5359aaa6d71345f22a3a952f13a6f6a56c0f8f3aedc0f8919028be5cfebd177",
    "Created": "2026-03-02T19:25:21.372001705+03:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "Gateway": "172.20.0.1"
        }
      ]
    }
  }
]

```

запускаем 3 контейнера в этой сети:

```

dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker run -d --name cont1 --network another-bridge anot
her-image:latest
f2050ec043bb45c16c87764415cdf45f9ec37b380e86ef1fd1157bdf03e1ba8
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker run -d --name cont2 --network another-bridge anot
her-image:latest
6005698db057d5336aae42809c19517fa70397fc6f5c0b111f1a55afb139b149
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker run -d --name cont3 --network another-bridge anot
her-image:latest
2215b0eebccd7ad603c6dc45b0cbd7438affb195fa94f4e6b3d0dcde731e1fc5

```

смотрим их ip адрес:

```

"Labels": {},
"Containers": {
  "2215b0eebccd7ad603c6dc45b0cbd7438affb195fa94f4e6b3d0dcde731e1fc5": {
    "Name": "cont3",
    "EndpointID": "23ec14a916626a62c9fdb7d6e05d35011bb72598cad4858a004c49545909155c",
    "MacAddress": "ca:37:2a:23:7e:12",
    "IPv4Address": "172.20.0.4/16",
    "IPv6Address": ""
  },
  "6005698db057d5336aae42809c19517fa70397fc6f5c0b111f1a55afb139b149": {
    "Name": "cont2",
    "EndpointID": "06f81255aa4a78dd00275aa37c93fd5de0ecf3d1b0265e3199e23ca0cab28ef1",
    "MacAddress": "f2:21:ec:be:cd:f8",
    "IPv4Address": "172.20.0.3/16",
    "IPv6Address": ""
  },
  "f2050ec043bb45c16c87764415cdf45f9ec37b380e86ef1fd1157bdf03e1ba8": {
    "Name": "cont1",
    "EndpointID": "fa1caa8332576150654f8f7580db00eaa6423fcab67afab02318c9530e5b0be1",
    "MacAddress": "7a:0b:3e:f2:3b:3f",
    "IPv4Address": "172.20.0.2/16",
    "IPv6Address": ""
  }
}
}

```

выключились).

пингуем их:

```

dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker exec -it cont1 sh
/ # ping -c 3 172.20.0.3
PING 172.20.0.3 (172.20.0.3): 56 data bytes
64 bytes from 172.20.0.3: seq=0 ttl=64 time=0.190 ms
64 bytes from 172.20.0.3: seq=1 ttl=64 time=0.109 ms
64 bytes from 172.20.0.3: seq=2 ttl=64 time=0.113 ms

--- 172.20.0.3 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.109/0.137/0.190 ms
/ # ping -c 3 172.20.0.2
PING 172.20.0.2 (172.20.0.2): 56 data bytes
64 bytes from 172.20.0.2: seq=0 ttl=64 time=0.069 ms
64 bytes from 172.20.0.2: seq=1 ttl=64 time=0.102 ms
64 bytes from 172.20.0.2: seq=2 ttl=64 time=0.104 ms

--- 172.20.0.2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.069/0.091/0.104 ms
/ # exit
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$

```

создаю overlay:

```

dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network create --driver overlay my-overlay
r46iuhbk0ytoicwc62pwtq0p6
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ network ls
network: команда не найдена
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
c5359aaa6d71        another-bridge      bridge              local
1818e6488f10        bridge              bridge              local
a748c7331900        docker_gwbridge     bridge              local
311a9e17d237        host                host                local
v4mwyoi2fhnq        ingress             overlay             swarm
8921180b0e36        my-bridge           bridge              local
r46iuhbk0yto        my-overlay          overlay             swarm
7f82da57511f        none                null                local
aa07e0fc2f6c        smarttrailsbackend_default bridge              local
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$

```

её информация:

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network inspect my-overlay
[
  {
    "Name": "my-overlay",
    "Id": "r46iuhbk0ytoicwc62pwtq0p6",
    "Created": "2026-03-02T16:40:39.221928712Z",
    "Scope": "swarm",
    "Driver": "overlay",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "10.0.1.0/24",
          "Gateway": "10.0.1.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
  }
]
```

создаём другую overlay:

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network create --driver overlay another-overlay
imjm37rtnhvlwgn71fyeur5pu
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network ls
NETWORK ID          NAME                DRIVER             SCOPE
c5359aaa6d71        another-bridge      bridge             local
imjm37rtnhvl        another-overlay     overlay           swarm
1818e6488f10        bridge             bridge             local
a748c7331900        docker_gwbridge     bridge            local
311a9e17d237        host               host              local
v4mwyoi2fhnq        ingress            overlay           swarm
8921180b0e36        my-bridge          bridge            local
r46iuhbk0yto        my-overlay         overlay           swarm
7f82da57511f        none              null              local
aa07e0fc2f6c        smarttrailsbackend_default bridge            local
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network inspect another-overlay
[
  {
    "Name": "another-overlay",
    "Id": "imjm37rtnhvlwgn71fyeur5pu",
    "Created": "2026-03-02T16:43:36.368695253Z",
    "Scope": "swarm",
  }
]
```

удаляем:

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network rm another-overlay
another-overlay
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network ls
NETWORK ID          NAME                DRIVER             SCOPE
c5359aaa6d71        another-bridge      bridge             local
1818e6488f10        bridge             bridge             local
a748c7331900        docker_gwbridge     bridge            local
311a9e17d237        host               host              local
v4mwyoi2fhnq        ingress            overlay           swarm
8921180b0e36        my-bridge          bridge            local
r46iuhbk0yto        my-overlay         overlay           swarm
7f82da57511f        none              null              local
aa07e0fc2f6c        smarttrailsbackend_default bridge            local
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$
```

попробуем создать host:

```
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$ docker network create --driver host my-host
Error response from daemon: only one instance of "host" network is allowed
dima@HP:~/try_2/453503_VASILENKO_7/IGI/LR2/test_2$
```